

PALANIMUTHU T

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PROFESSIONAL SUMMARY

Data-focused Associate Software Engineer with 1+ year of professional experience working with Python, SQL, data processing, reporting, automation, and production application workflows. Hands-on experience developing an end-to-end ETL pipeline using Python, Pandas, JSON, SQLite, SQL, and relational data modelling, including data cleaning, normalization, validation, and star-schema design. Professional experience includes building and maintaining a 22-page Power BI MIS reporting solution with scheduled data refreshes from Azure Cosmos DB. B.Tech graduate in Artificial Intelligence and Data Science with a strong interest in data engineering and scalable data workflows.

EXPERIENCE

Maadhyamik Technologies Private Limited

Associate Software Engineer | Salem, Tamil Nadu | July 2025– Present

- Built and maintained a **22-page Power BI MIS reporting solution** for internal business and application-performance monitoring.
- Worked with structured operational data to create KPI-focused reports, interactive filters, drill-downs, and analytical views across dimensions such as age, subscription, and country.
- Supported **twice-daily scheduled data refreshes from Azure Cosmos DB** and monitored reporting data for consistency and availability.
- Developed Python/Pandas workflows for **data cleaning, transformation, normalization, and validation** across application and reporting-related processes.
- Created Python automation scripts for Excel-based data processing and repetitive content/data workflows.
- Worked with SQL, Python, Excel, and Power BI as part of data processing, reporting, and production workflows.
- Collaborated on data-driven features and production workflows for a live mobile application serving **10K+ users**.
- Used APIs and automation tools including Gemini API and Azure Cognitive Services in application and data-related workflows.

PROJECTS

IPL Data Analytics ETL Pipeline

Python, Pandas, SQL, SQLite, JSON, ETL, Data Modelling

- Developed an end-to-end **ETL pipeline** using cricket match data sourced from Cricsheet JSON files.
- Extracted and processed raw JSON data using Python and Pandas.
- Performed data cleaning, transformation, normalization, and validation before loading structured data into SQLite.
- Designed a **relational star schema** using fact and dimension tables for matches, players, teams, and related entities.
- Used SQL to perform analytical queries and generate structured insights from the transformed dataset.
- Built the project to demonstrate an end-to-end data workflow from **raw data ingestion** → **transformation** → **data modelling** → **storage** → **analytics**.

Hospitality Data Analysis

Python, SQL, Power BI, Excel, Pandas

- Analysed **119K+ hotel booking records** using Python and SQL.
- Performed data cleaning and transformation to prepare booking data for analysis.
- Developed a Power BI dashboard covering ADR, cancellation rate, booking trends, average stay, repeat guests, hotel type, country, market segment, and booking channel.
- Used KPI analysis and dimensional breakdowns to identify trends and patterns in hospitality data.

Sales & Profitability Analysis

Excel, PivotTables, PivotCharts, Slicers

- Analysed sales data representing **₹26.68 Cr in sales and ₹5.00 Cr in profit**.
- Built an interactive Excel dashboard covering revenue, profit, margin, satisfaction, and return-rate KPIs.
- Used PivotTables, PivotCharts, and slicers to analyse performance across regions, payment methods, and product categories.

TECHNICAL SKILLS

- Programming:** Python, SQL
- Data Engineering & ETL:** ETL, Data Cleaning, Data Transformation, Data Validation, Data Normalization, Star Schema
- SQL:** Joins, Aggregations, Filtering, Grouping
- Python & Data Processing:** Pandas, NumPy, OpenPyXL, JSON
- Databases:** SQLite, Azure Cosmos DB
- Data Analytics:** Data Cleaning, Transformation, Validation, ETL, Data Modelling, KPI Analysis
- BI & Analytics:** Power BI, DAX, Power Query, Excel, PivotTables, PivotCharts, KPI Analysis
- Tools:** Jupyter Notebook, VS Code, Azure Cognitive Services, Gemini API

EDUCATION

Bachelor of Technology – Artificial Intelligence and Data Science

Sona College of Technology, Salem

2021 – 2025 | **CGPA: 8.04/10**

ACHIEVEMENTS

- Contributed to production features released in a live mobile application.
- NCC C Certificate – A Grade.